



Chapter 1

Introduction

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CS-241



Recommended books

1. Data Communications and Networking, 4th Edition. Behrouz A. Forouzan
2. Computer_Networking_A_Top-Down_Approach (7TH ED)

1-1 DATA COMMUNICATIONS

- The term *telecommunication* means communicating over a distance in other words, sending and receiving messages when the sender and receiver are far apart.
- The word *data* means information that can be in any form (like *text, numbers, images, or sounds*) — as long as both parties (the sender and receiver) agree on how that information is represented and understood.
- So, *data Communication* refers to the exchange of data between two electronic devices (like computers or phones) through a transmission medium, such as a wire cable, fiber optics, or even wireless signals.

Topics discussed in this section:

Components

Data Representation

Data Flow



Four Fundamental Characteristics

1. **Delivery:** The data must be delivered to the correct destination.

Example: A message sent from your computer should reach the intended receiver not anyone else.

2. **Accuracy:** The data must be delivered correctly, without any alteration or error.

Example: If you send the number “250”, it should not arrive as “520”.

3. **Timeliness:** Data must be delivered on time. Delayed data is often useless (especially in real-time systems like video calls or stock trading).

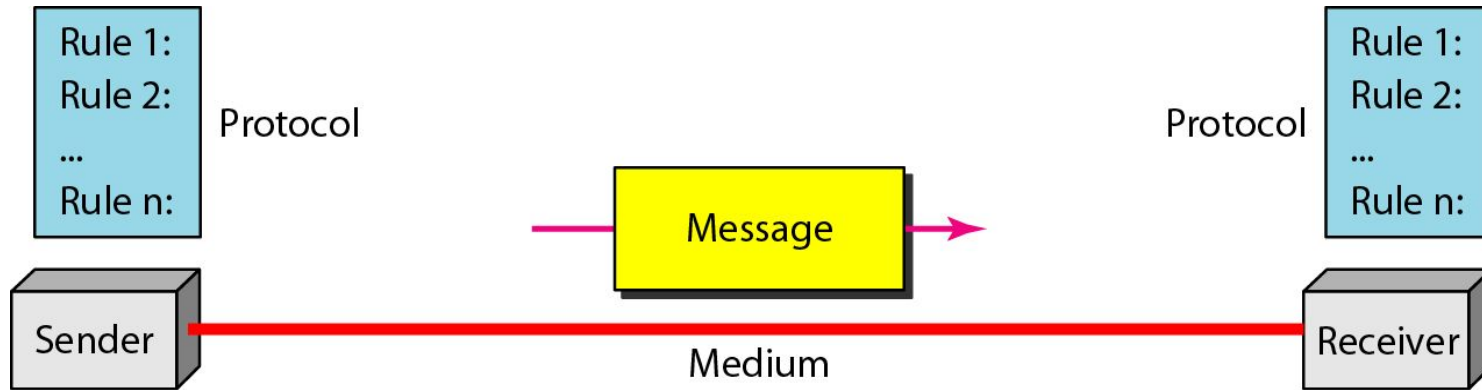
Example: In a video call, if the voice arrives 3 seconds late, it affects communication that’s a timeliness issue.

4. **Jitter:** It is the variation in packet arrival time. In simple terms, jitter means uneven delay in data delivery.

Example:

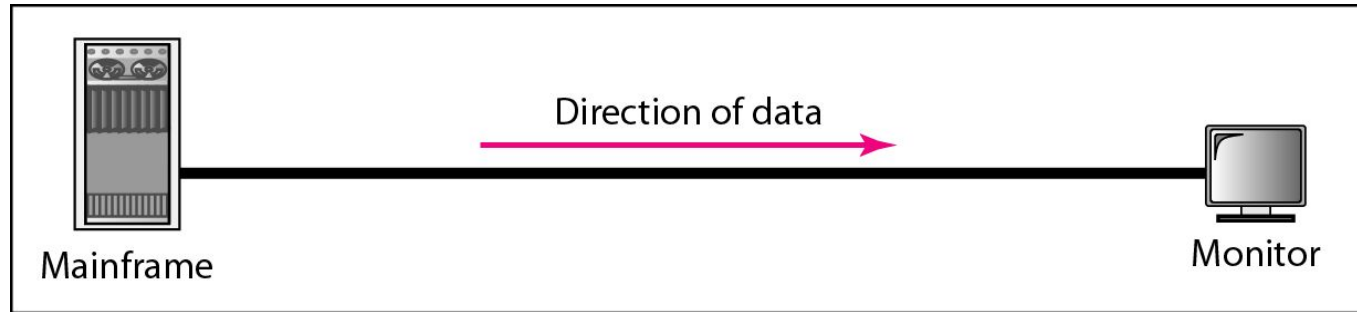
In a video call, if one voice packet arrives in 1 ms, the next in 10 ms, and another in 5 ms that’s jitter

Figure 1.1 *Five components of data communication*

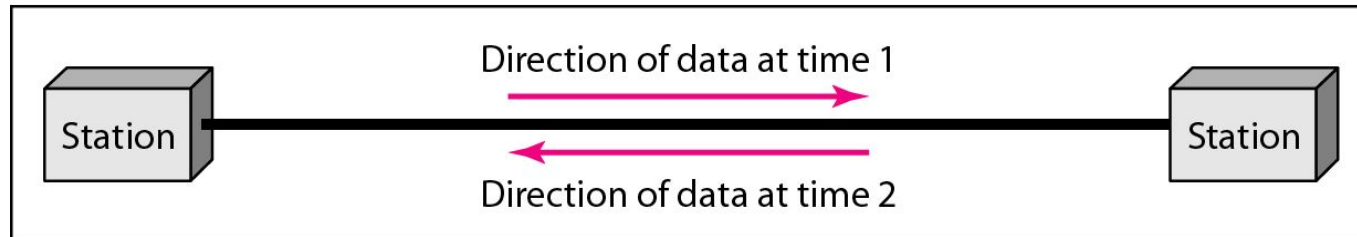


Component	Description	Example
Message	Data to be sent	Text, audio, video
Sender	Source of data	Computer, phone
Receiver	Destination of data	Server, printer
Transmission Medium	Path of data	Cable, Wi-Fi
Protocol	Rules for communication	TCP/IP, HTTP

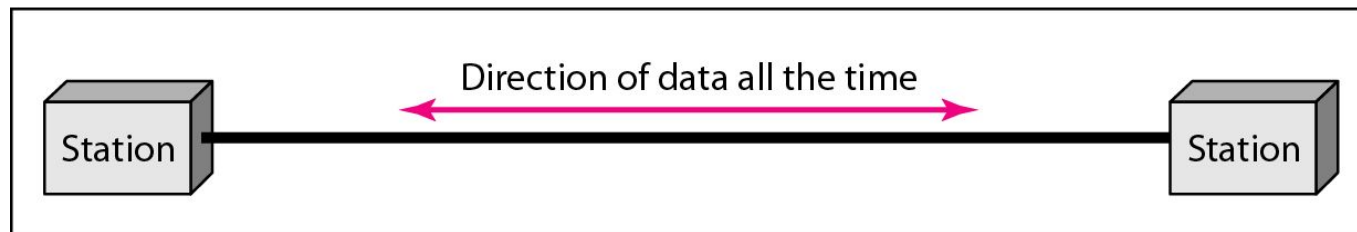
Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



a. Simplex



b. Half-duplex



c. Full-duplex

Data Transmission Modes

Simplex Mode:

- Data flows in only one direction. One device is sender, the other is receiver they cannot switch roles.
- **Example:** Keyboard → Monitor. Printer receiving data from a computer.

Half-Duplex Mode

- Data can flow in both directions, but not at the same time.
- Each device can send or receive, but only one at a time.
- **Example:** Walkie-talkie communication (you talk, then listen)

Full-Duplex Mode

- Data can flow in both directions simultaneously. Both devices can send and receive at the same time.
- **Example:** Mobile phone calls, Internet communication

1-2 NETWORKS

- A network is a group of two or more devices (computers, printers, servers, etc.) that are connected by communication links to exchange data and resources.
- **Example:** When your computer connects to Wi-Fi, it becomes part of a network that may include:
 - Your phone
 - Your printer
 - The internet router
- All of them can send and receive information through this network.
- **Main Purpose of a Network:**
 1. **Communication** – Email, chatting, video calls
 2. **Resource sharing** – Sharing printers, files, and storage
 3. **Data sharing** – Accessing shared databases or websites
 4. **Remote access** – Connecting to devices or systems from anywhere.

1-2 NETWORKS

*A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.*

Topics discussed in this section:

Distributed Processing

Network Criteria

Physical Structures

Network Models

Categories of Networks

Interconnection of Networks: Internetwork



Distributed Processing

- Most networks use distributed processing, in which a task is divided among multiple computers. Instead of one single large machine being responsible for all aspects of a process, separate computers (usually a personal computer or workstation) handle a subset.
- **Real-Life Example:**
- Imagine you have to check 1,000 exam papers. Instead of doing it alone:
 - You give 300 to your friend
 - 300 to another friend
 - You check 400 yourself
- When everyone finishes, you combine all the results. That's distributed processing the same idea used in computer networks.

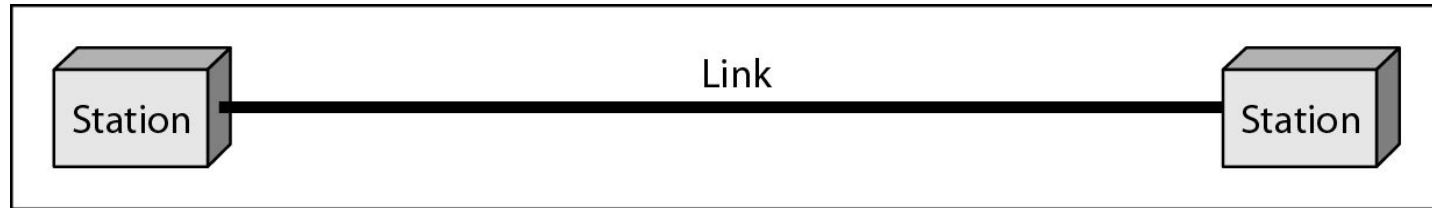


Network Criteria

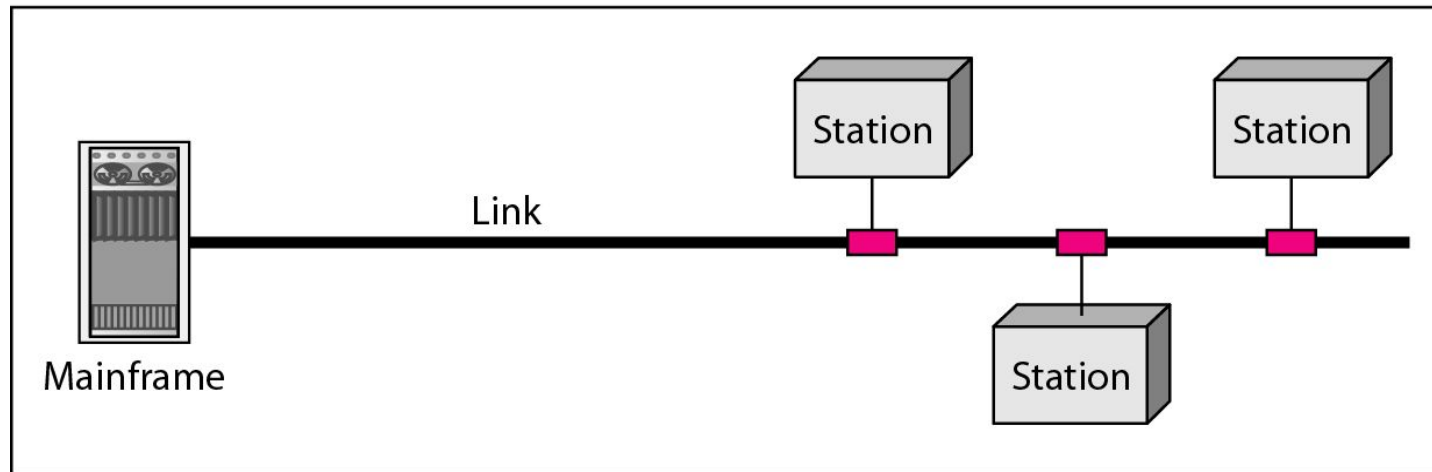
- 1. Performance** {throughput, delay}
- 2. Reliability** {robust network}
- 3. Security**

Criteria	Meaning	Example
Performance	How fast the network works	Internet speed while loading YouTube
Reliability	How stable the network is	Wi-Fi not disconnecting again and again
Security	How safe the data is	Protection from hackers

Figure 1.3 *Types of connections: point-to-point and multipoint*



a. Point-to-point



b. Multipoint



Types of Networks

1. PAN – Personal Area Network:

- **Range:** A few meters
- **Used for:** Connecting personal devices around one person
- **Examples:**
 - Bluetooth between mobile and earbuds
 - Laptop connected to a smartphone hotspot

2. LAN – Local Area Network:

- **Range:** Within a building or small area (like a school, office, or home)
- **Used for:** Sharing data, printers, and internet within one location
- **Examples:**
 - Computer lab in a school
 - Office network connected to a single router

3. MAN – Metropolitan Area Network:

- **Range:** Covers a city or a large campus
- **Used for:** Connecting multiple LANs across a city
- **Examples:**
 - Network connecting university campuses across a city
 - Cable TV or citywide Wi-Fi network



Types of Networks

4. WAN – Wide Area Network

- **Range:** Covers countries or continents
- **Used for:** Connecting multiple LANs and MANs worldwide
- **Examples:**
 - The Internet (biggest WAN)
 - Multinational company networks

Comparison Table:

Type	Full Form	Range	Example
PAN	Personal Area Network	A few meters	Bluetooth, Hotspot
LAN	Local Area Network	One building	School or office network
MAN	Metropolitan Area Network	City	Cable TV, city Wi-Fi
WAN	Wide Area Network	Country or world	Internet



Network Topology

- Network topology means the way computers and devices are connected in a network or the arrangement (shape or layout) of the network.

Figure 1.4 *Categories of topology*

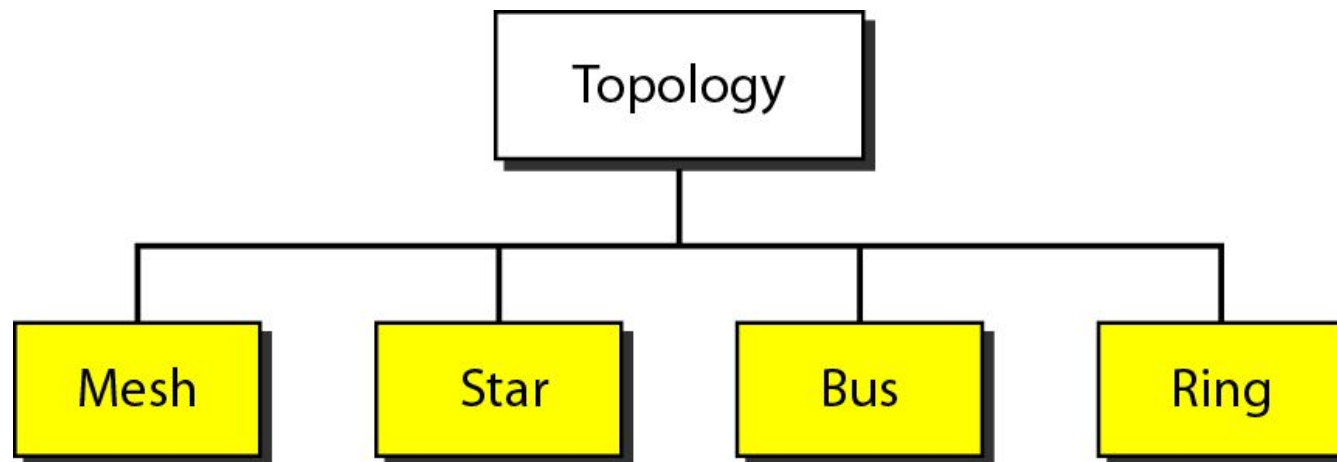
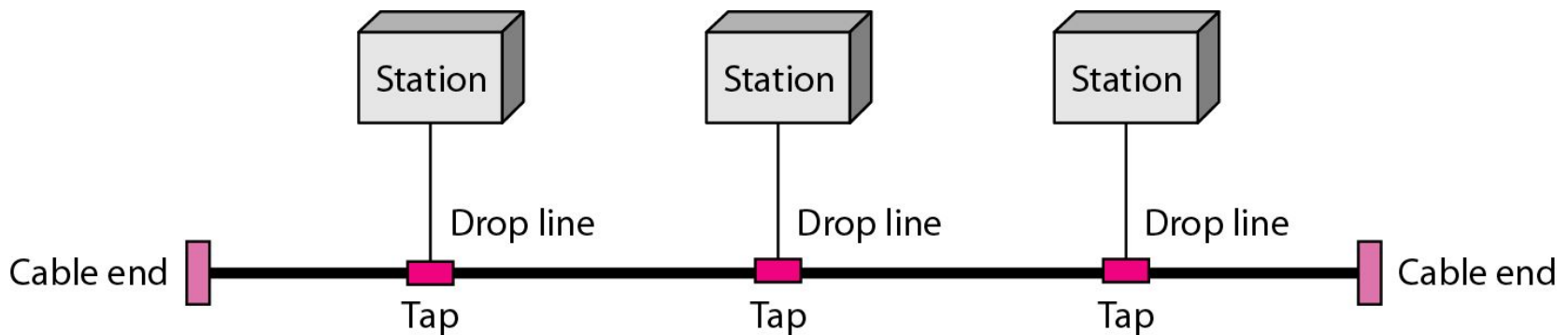


Figure 1.5 *A bus topology connecting three stations*

- Bus topology is a network setup where all computers and devices are connected to a single main cable, called the **bus** or **backbone**.
- Think of it like a bus route all bus stops (computers) are connected to the same road (main cable).
When one computer sends data, it travels along that main cable and can be received by any other computer on it.



Bus topology

Advantages:

1. Easy to install and set up.
2. Requires less cable, so it's cheap.
3. Works well for small networks.

Disadvantages:

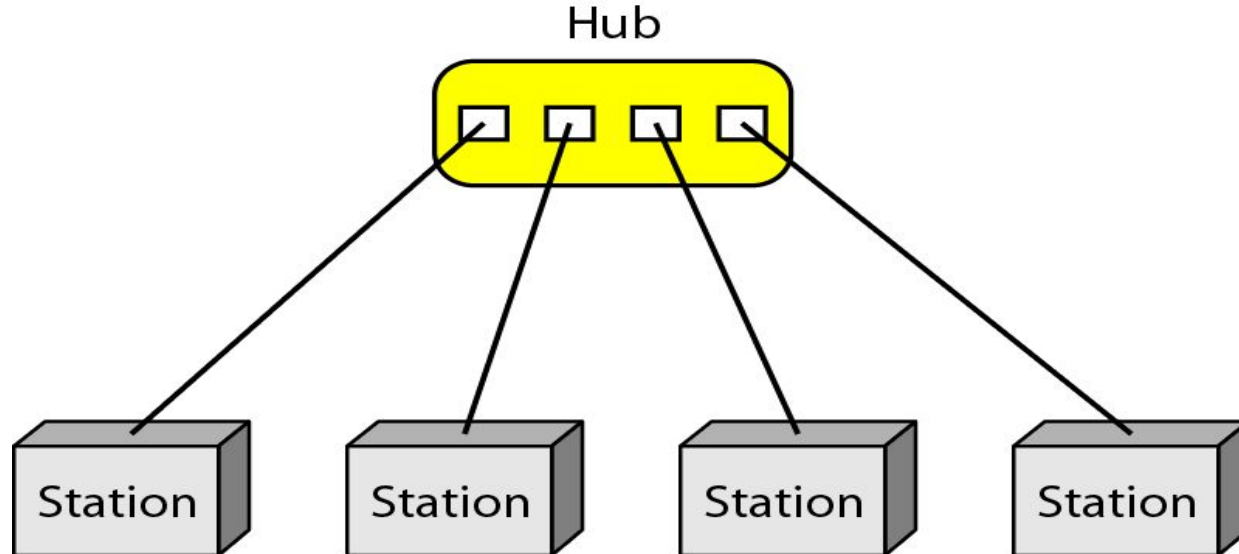
1. If the main cable fails, the whole network stops.
2. Difficult to troubleshoot when there's a problem.
3. More devices = more data collisions (network slows down).
4. Not good for large networks.

Examples:

- Old LAN networks in offices and schools.
- Early Ethernet networks used bus topology.

Figure 1.6 *A star topology connecting four stations*

- Star topology is a network setup where all computers are connected to a central device, like a hub, switch, or router.
- Each computer has its own separate cable that connects directly to the central device.
- Think of a star shape the center is the hub/switch, and each point of the star is a computer connected to it.



Star

Topology

Advantages:

1. Easy to install and manage simple to add or remove computers.
2. If one cable fails, only that device is affected (network continues working).
3. Fast performance because each computer has a dedicated connection to the hub.
4. Easy to detect and fix faults.

Disadvantages:

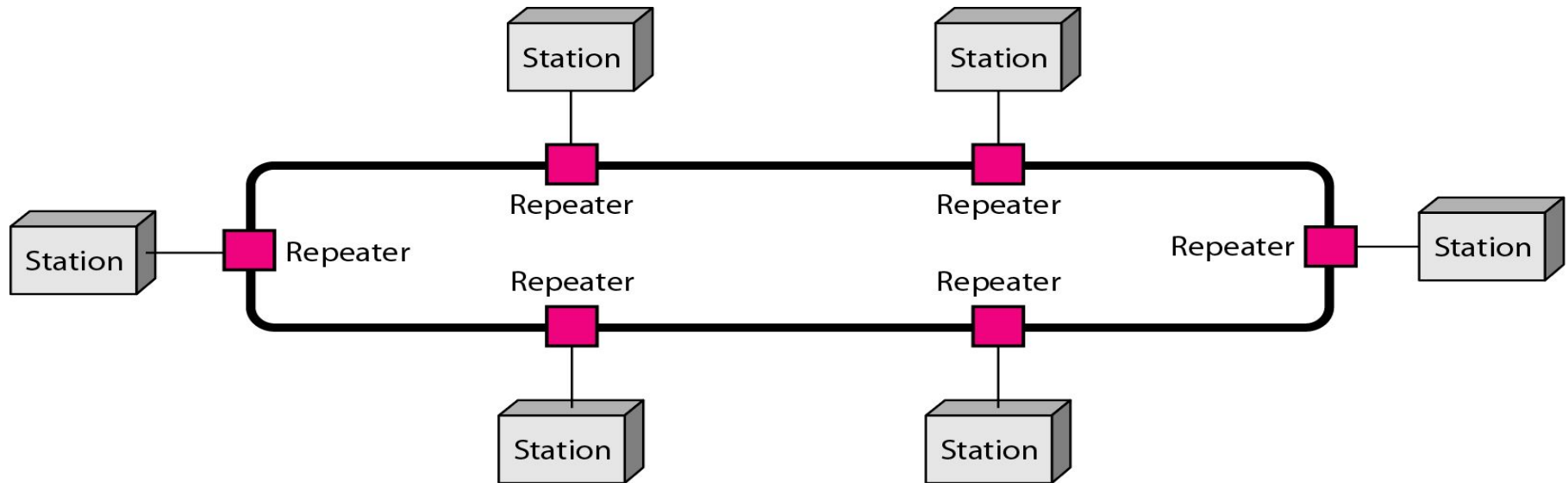
1. More cables needed → higher cost than bus topology.
2. If the central hub or switch fails, the entire network stops.
3. Performance depends on the capacity of the central device.

Examples:

- Most modern LANs (Local Area Networks) use star topology.
- Wi-Fi routers in homes or offices act like the central hub in a star network.

Figure 1.7 *A ring topology connecting six stations*

- Ring topology is a network setup where each computer is connected to exactly two other computers, forming a closed loop (ring).
- Data travels in one direction (or sometimes both) around the ring until it reaches its destination.
- Think of a circle where each computer passes a message to the next one like students sitting in a circle passing a note around.



Ring Topology

Advantages:

- 1.No data collision, because data moves in one direction.
- 2.Equal access for all computers (each gets a turn to send data).
- 3.Works well for small to medium networks.

Disadvantages:

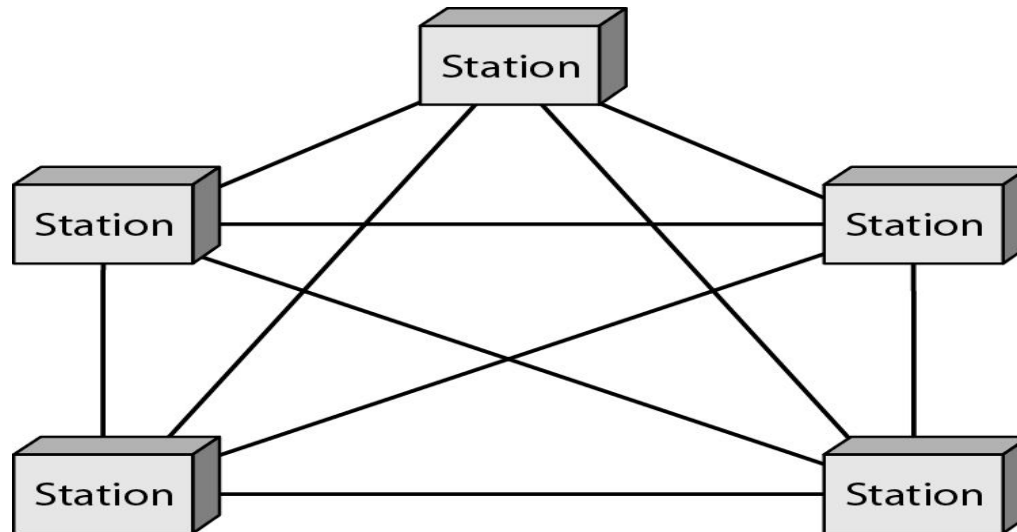
- 1.If one computer or cable fails, the whole network can stop (unless it's a dual ring).
- 2.Adding or removing devices is difficult.
- 3.Troubleshooting is harder compared to star topology.
- 4.Slower than star topology when many devices are connected.

Examples:

- Used in some LANs and token ring networks (older IBM networks).
- Some FDDI (Fiber Distributed Data Interface) networks also use ring topology.

Figure 1.8 *A fully connected mesh topology (five devices)*

- Mesh topology is a network setup where every computer or device is connected to every other device directly.
- This means there are multiple paths for data to travel if one path fails, data can take another route.
- Think of it like each friend in a group having everyone else's phone number if one phone stops working, they can still contact others directly.



Mesh Topology

Advantages:

1. Very reliable — if one link fails, data can use another path.
2. No data traffic problems — multiple paths available.
3. High security — data has dedicated routes.
4. Fast communication between devices.

Disadvantages:

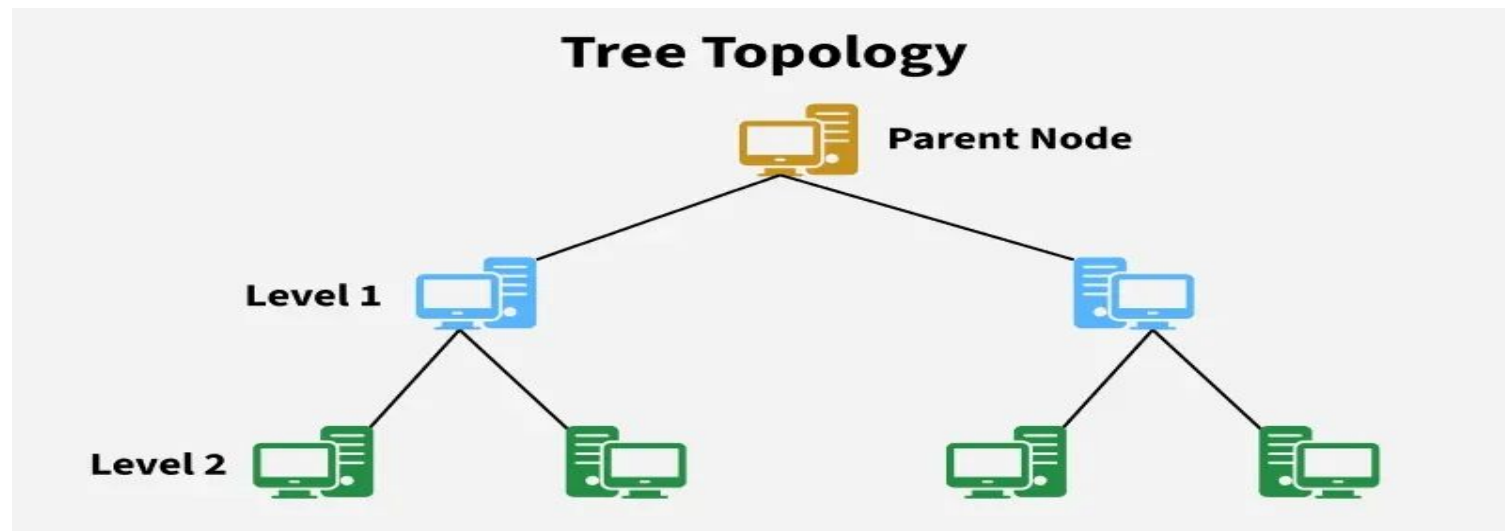
1. Expensive — needs a lot of cables and network ports.
2. Difficult to install and maintain in large networks.
3. Complex structure — not practical for small organizations.

Examples:

- Used in WANs (Wide Area Networks) and wireless networks.
- Some internet backbone connections use mesh topology for reliability..

Figure 1.9 *A tree topology (seven devices)*

- Tree topology is a combination of bus and star topologies. It has a root node (main hub) that connects to other nodes in a hierarchical or branching structure like a tree with branches.
- The topmost node is called the root. It connects to one or more secondary nodes (branches).
- Each secondary node can connect to its own sub-branches (child nodes).



Mesh Topology

Example:

- Imagine a university network:
- **Root node:** Main server in the IT department
- **Branches:** Departmental switches (e.g., CS, Physics, Math)
- **Leaves:** Computers and printers in each department
- **Advantages:**

1. Scalable: Easy to add new devices or branches.
 2. Fault Isolation: Problems in one branch don't affect others.
- Hierarchical management: Makes network control organized.

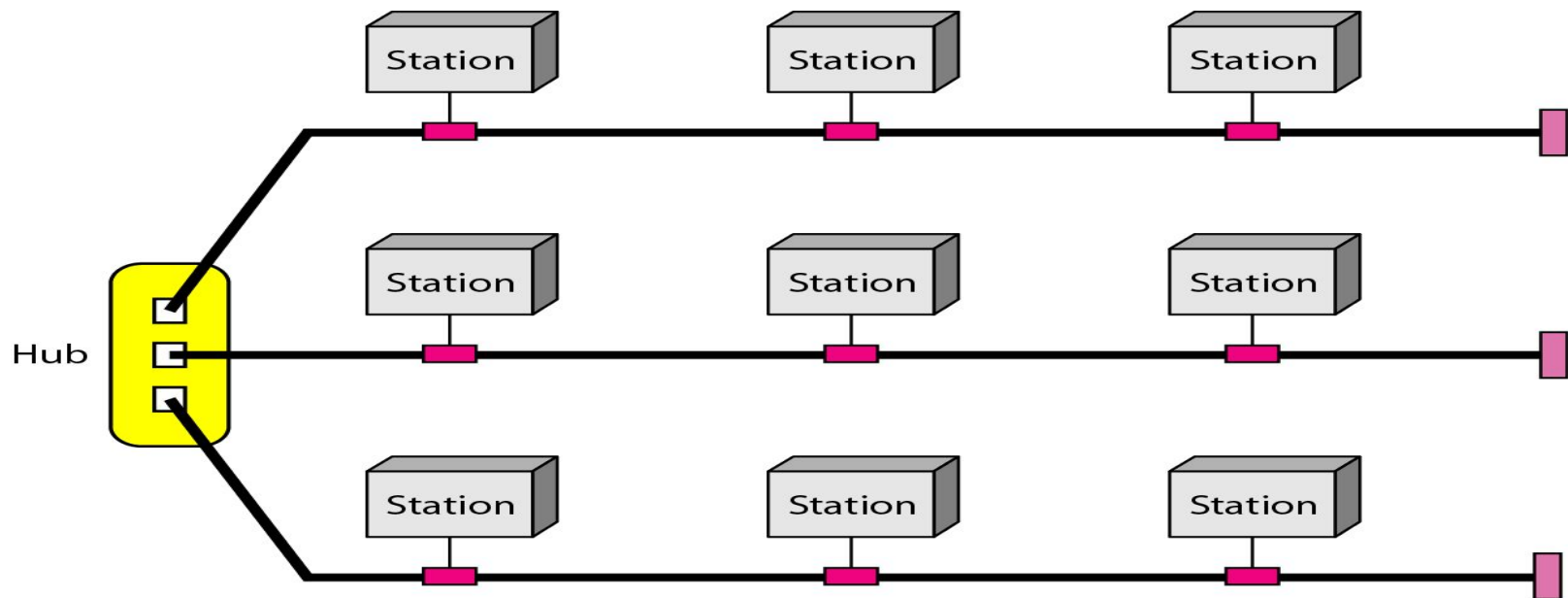
Disadvantages:

- **Expensive:** Requires more cables and network devices.
- **Failure at Root:** If the root hub fails, the whole network can go down.
- **Complex setup:** More complicated to design and maintain.

Figure 1.10 *A hybrid topology: a star backbone with three bus networks*

A Hybrid Topology is a combination of two or more different network topologies such as star, bus, ring, or mesh to make a network that uses the advantages of each.

Think of hybrid topology like mixing different shapes — for example, a star inside a ring or a bus connecting multiple star networks



Hybrid Topology

Advantages:

1. Flexible — you can design it based on your needs.
2. Scalable — easy to expand the network.
3. Reliable — if one part fails, others can still work.
4. Efficient — combines the best features of multiple topologies.

Disadvantages:

1. Expensive — more cables and devices are needed.
 2. Complex design and maintenance.
 3. If the main connecting device or backbone fails, it can affect the whole network.
- Large organizations and universities often use hybrid topology.

Example:

A university with several departments, each using a star topology, all connected through a bus or ring backbone.

Formulation of global network

- The Internet is a Combination of Topologies: The global Internet isn't based on a single topology it's a hybrid (mixed) network made up of many different topologies working together at different levels.

Local Area Networks (LANs): At the local level (like in homes, offices, schools):

- Most use Star Topology
 - All devices connect to a central device (router/switch).
 - **Example:** Your Wi-Fi router connects your phone, laptop, and printer.
- Sometimes Bus or Ring topologies are used in small or legacy systems.

Metropolitan Area Networks (MANs): In cities, networks connect multiple LANs.

- Commonly use Ring or Mesh topologies for reliability.
- Data travels in a loop or along multiple paths between routers.
- **Wide Area Networks (WANs):** At the global level, networks connect countries and continents. Use **Mesh Topology** multiple redundant paths between routers. Undersea cables, satellites, and backbone routers are all interconnected. If one route fails, data automatically takes another.

Formulation of global network

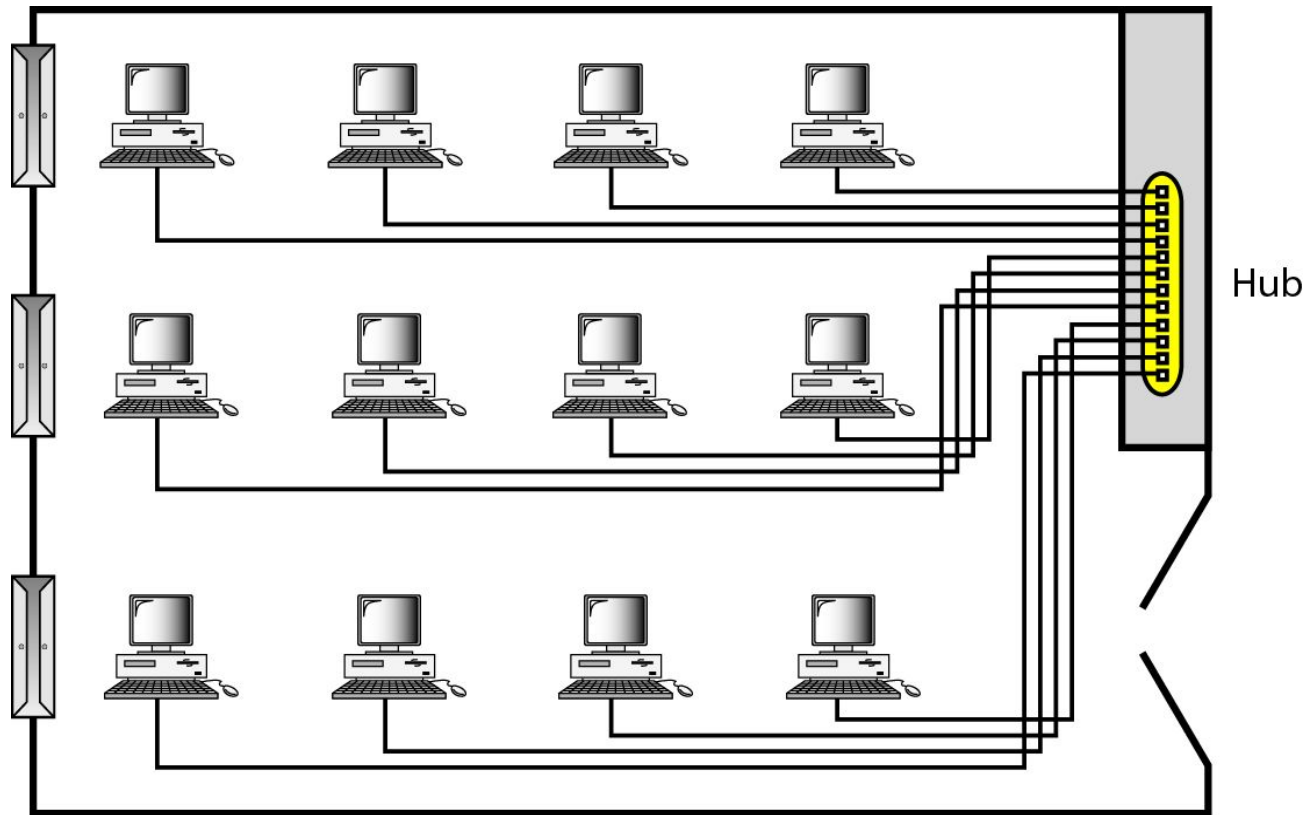
Overall Structure — A Hybrid

So the worldwide network (Internet) is basically a hybrid topology, made up of:

- Star** (in homes and offices)
- Ring** (in city or regional links)
- Mesh** (in global backbone connections)

Network Level	Common Topology	Example
Home/Office (LAN)	Star	Wi-Fi router and devices
City (MAN)	Ring / Mesh	Fiber network around a city
Global (WAN/Internet)	Mesh	Internet backbone (ISPs, satellites)
Overall	Hybrid	Combination of all

Figure 1.11 *An isolated LAN connecting 12 computers to a hub in a closet*



1-3 THE INTERNET

*The **Internet** has completely changed many parts of our daily lives. It has transformed how we do business, learn, and communicate, as well as how we spend our free time. The Internet is a powerful communication system that provides access to a huge amount of information. It brings knowledge from around the world right to our fingertips and organizes it so we can use it easily.*

Topics discussed in this section:

A Brief History

1-4 PROTOCOLS AND STANDARDS

*In this section, we define two widely used terms: **protocols** and **standards**. First, we define protocol, which is synonymous with rule. Then we discuss standards, which are agreed-upon rules.*

Topics discussed in this section:

Protocols

Standards

Standards Organizations

Internet Standards

Thank you!

